Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

A. follicle-stimulating hormone

B. oxytocin

C. pituitary gland

D. hypothalamus

E. gonadotropin-releasing hormone

F. luteinizing hormone

G. testosterone

Reasoning + Answer:

<think>

In the male reproductive endocrine feedback loop, the hypothalamus synthesizes and releases gonadotropin-releasing hormone (GnRH) to the anterior pituitary. This stimulates release of follicle-stimulating hormone (FSH) and luteinizing hormone (LH). In turn, Leydig cells in the testis are LH-dependent, synthesizing testosterone. Elevated testosterone, through peripheral tissues and the brain, inhibits hypothalamic GnRH secretion and pituitary FSH/LH release, forming a negative feedback loop.

The component(s) that do not belong in this feedback loop are:

- FSH: Produced by the pituitary, it is part of the feedback loop stimulating testicular testosterone production. But it is not directly released by the hypothalamus in response to testosterone, so it is not included in the feedback loop as a regulator of its own secretion.
- Oxytocin: A vasodilatory hormone, it does not participate in the endocrine regulation of sex hormone secretion in this loop.
- Testosterone: An endocrine product, not a pituitary hormone or hypothalamic signal, so it is not part of the feedforward/feedback mechanism that regulates gonadotropin secretion.
- Pituitary gland: Secretes FSH and LH in response to hypothalamic GnRH; it is downstream of the hypothalamic feedback signal, not part of the direct hypothalamic âĢĵpituitary loop itself.
- Hypothalamus: Secretes GnRH to stimulate pituitary FSH/LH; it is central to the feedback mechanism.

Therefore, the element that does not belong in this loop is testosterone.

Correct answer: G. testosterone

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<answer>

G. testosterone

</answer>